

Athena-KB

A MULTI-SOURCE RAG PLATFORM FOR OFFENSIVE SECURITY KNOWLEDGE MANAGEMENT

AVA NEARY

BSC (HONS) APPLIED COMPUTING

SUPERVISOR: DR. BERNARD BUTLER

01

ABSTRACT

Athena-KB is a proof-of-concept implementation of a fully self-hosted Retrieval-Augmented Generation (RAG) system. It includes an external ingestion component and a self-managed AI inference stack, built entirely on open-source software. It provides full end-to-end control over data, ingestion and retrieval through configurable pipelines, allowing operators to curate, manage and query knowledge bases from both internal and external sources, and receive transparent, source-backed answers.

02 MOTIVATION

- Cybersecurity evolves rapidly, making knowledge maintenance a constant challenge.
- Entry-mid level red team operators lack access to deep, tactical knowledge that seasoned operators have curated over years.
- Commercial AI solutions can't be trusted with sensitive data.
- LLMs can tailor responses to a user's experience level, making knowledge accessible across all skill levels.

03 KEY FEATURES

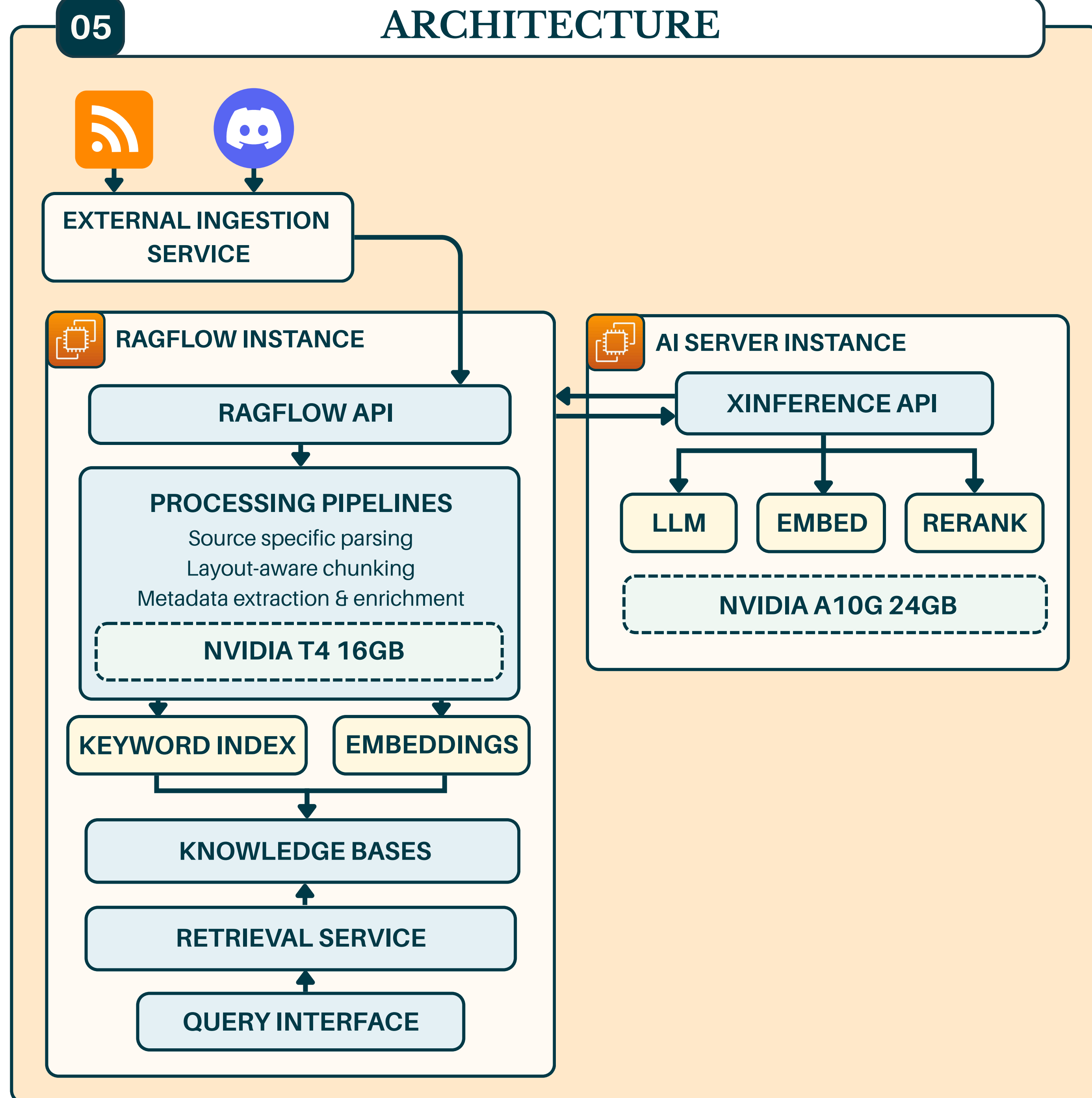
- Scoped knowledge bases.
- External ingestion initially supporting RSS and Discord, with most file types accepted via direct upload
- Ingestion pipelines with metadata enrichment for improved retrieval.
- AWS infrastructure provisioned through IaC.
- Web UI for knowledge base management, pipeline configuration and querying.

04 TECHNOLOGIES



05

ARCHITECTURE



06

QUERY FLOW

